

Tutorial 7B Solutions

1. Find a vector equation of the line that
- (a) is parallel to $-2\mathbf{i} + 3\mathbf{j}$ and passes through the origin,
 - (b) passes through the points $(2, 1, 0)$ and $(1, 1, -3)$,
 - (c) passes through $(4, -1, 2)$ and is parallel to the x -axis.

Solution

(a) Equation is $\underline{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}, \lambda \in \mathbb{R}.$

(b) $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$

Equation is $\underline{r} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}.$

(c) Equation is $\underline{r} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \lambda \in \mathbb{R}.$

Vector equation of a line is given by

$$l : \underline{r} = \underline{a} + \lambda \underline{d}, \lambda \in \mathbb{R},$$

where $\underline{a}$ is position vector of a fixed point on the line and $\underline{d}$ is the direction vector parallel to the line.

To find vector equation of a line, use the position vectors of 2 points to form the direction vector, $\underline{d}$.

Remember to represent a vector quantity in written work by a letter in lowercase with a tilde below the letter, e.g. $\underline{a}$, $\underline{k}$, $\underline{p}$.

2. Convert the following Cartesian equations of straight lines to vector equations.

(a) $\frac{1-x}{5} = \frac{y}{5} = z-1$

(b) $\frac{2x+1}{2} = y, z=5$

Does (1, 0, 1) lie on the lines above?

Solution

(a) Equation is $\underline{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 5 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}.$

(b) Equation is $\underline{r} = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \mu \in \mathbb{R}.$

Cartesian/vector equations of a line:

$$\frac{x-a_1}{d_1} = \frac{y-a_2}{d_2} = \frac{z-a_3}{d_3}$$

$$\Leftrightarrow \underline{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \mu \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

When $\lambda = 0$, $\underline{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, so (1, 0, 1) lies on the line in (a).

Equation in (b) has $z = 5$. Since $z = 1$ in (1, 0, 1), it does not lie on the line in (b).

3. [A Level/N00/PI/Q3]

(a) Equations of two lines are given below:

$$l_1 : \mathbf{r} = -2\mathbf{i} + \mathbf{j} + 9\mathbf{k} + \lambda(2\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}) \text{ where } \lambda \in \mathbb{R},$$

$$l_2 : \mathbf{r} = 11\mathbf{i} + 8\mathbf{j} + 3\mathbf{k} + \mu(3\mathbf{i} - \mathbf{j} + 5\mathbf{k}) \text{ where } \mu \in \mathbb{R}.$$

Determine if the lines are parallel, intersecting or skew, giving a reason for your answer.

Solution

We have $\begin{pmatrix} 2 \\ 5 \\ 4 \end{pmatrix} \neq k \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ for any k , hence the lines are not parallel. Can also use GC to solve for λ and μ .

Assuming the two lines intersect:

$$-2 + 2\lambda = 11 + 3\mu \quad \text{--- (1)}$$

$$1 + 5\lambda = 8 - \mu \quad \text{--- (2)}$$

$$9 + 4\lambda = 3 + 5\mu \quad \text{--- (3)}$$

$$3 \times (2) \Rightarrow 3 + 15\lambda = 24 - 3\mu \quad \text{--- (4)}$$

$$(1) + (4): \quad 1 + 17\lambda = 35$$

$$\lambda = 2, \quad \mu = -3$$

Substitute $\lambda = 2, \mu = -3$ into (3): $\text{LHS} = 17 \neq -12 = \text{RHS}$

$\therefore$ the 2 lines do not intersect.

Since the 2 lines are non-parallel and non-intersecting, therefore they are skew lines.

